INTERIM AMENDMENT

TO

FEDERAL SPECIFICATION

PAINT, STYRENE-BUTADIENE SOLVENT TYPE, WHITE
(FOR EXTERIOR MASONRY)

This interim amendment was developed by the Federal Supply Service, General Services Administration, Washington, D. C. 20406, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

The General Services Administration has authorized the use of this interim amendment as a valid exception to Federal Specification TT-P-97D, dated January 25, 1968.

PAGE 1

Under Federal Specifications, delete the following:

TT-P-442 - Pigment, Titanium Dioxide, (For Protective Coatings).

PAGE 2

Before 3. REQUIREMENTS. Add the following:

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D-476 - Titanium Dioxide Pigments

Paragraph 3.1, line 4. Add "The solvents or solvent system used in the process shall comply with Air Pollution Regulations "Rule 66"1/. A certificate of compliance to this effect is necessary."

TABLE 1, line 13 and bottom of page. Change footnote1/ to footnote2/.

Bottom of page. Insert new footnote1/ as follows:

"1/Information on "Rule 66" may be obtained from Air Pollution Control District, County of Los Angeles, Los Angeles, California 90001".

PAGE 3

Table II, line 1, change "TT-P-442" to "ASTM D-476".

PAGE 4

Paragraph 4.4.6 - 4.4.7 (inc.). Delete in entirety and substitute:

4.4.6 Determination of styrene-butadiene resin.

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Apparatus: Supercentrifuge
Centrifuge
Double beam grating infrared recording spectrophotometer (Spectral resolution 3 cm or better over entire range at slow scan)
Rock salt plates for use in spectrophotometer
Vacuum oven
Drying oven
Soxhlet extraction apparatus-500-ml erlenmeyer flasks - 24/40 neck
Hot plates

4.4.6.1 Determination of chlorinated paraffins and additives (n-pentane extractables).

Weigh, by difference to the nearest mg., 50 to 55g of well-mixed sample, using a weighing bottle and add to 100g of sodium chloride C.P. in a 500-ml glazed porcelain casserole. Mix well and evaporate most of the solvent by heating the casserole on top of the steam bath (not directly over an opening) for 6 hours. Stir mixture occasionally during evaporation. By using an infrared lamp in addition to the steam bath, the solvent evaporation time can be reduced from 6 to 2 hours. Remove the mixture from the casserole while still slightly damp and grind as finely as possible in a mortar. Dry in the casserole overnight in a regular drying oven at - 70°C. There should be no odor of solvent. Regrind to a powder as fine as salt and transfer to a Soxhlet thimble (43 x 123 mm) and cover with a plug of glass wool. Extract with 400 ml of technical grade n-pentane, under vigorous reflux for eight hours, or if more convenient, let run overnight (see note). Use a Soxhlet extraction apparatus equipped with a calcium chloride tube and five Berl saddles as boiling stones in the 500 ml erlenmeyer flask.

At the conclusion of the extraction period, remove the extraction thimble and rinse with n-pentane. Rinse the Soxhlet with n-pentane. Transfer the warm pale yellow n-pentane extract and rinsings to a tared 400-ml beaker containing three Berl saddles. Rinse the extraction flask with n-pentane and add to the 400-ml beaker. The film remaining on the bottom and sides of the flask is low molecular weight styrene-butadiene resin and is not to be combined with the n-pentane extract which contains the chlorinated paraffins. Evaporate the n-pentane extract on a cooler portion of the steam bath. When the n-pentane has been evaporated, leave the beaker directly over the opening at full steam to remove any traces of solvent. Dry in a vacuum oven to constant weight at 60 - 70°C. This requires overnight (16 hrs.) drying. Weigh beakers and calculate weight of n-pentane extract.

Notes

1. The salt-paint mixture is evaporated on top of the steam bath to prevent any decomposition of the styrene butadiene resin. It was found that white paints containing styrene resins became gray when placed in a casserole directly over the steam opening, owing to decomposition. If, on the other hand, the same paint was evaporated on top of the steam bath, it remained white. Temperature measurements showed 73°C on top and 100°C below the steam bath opening.

2. The purpose of evaporation is to eliminate the original paint solvents. These must be eliminated in order to reduce as much as possible the solubility of the styrene-butadiene resin in the n-pentane.

3. The salt-paint mixture should be removed from the casserole while still damp. If allowed to dry, it acquires a rock-like hardness, thus making removal and grinding laborious and time-consuming.

4. It was found that 98 percent of the n-pentane extract obtained in a 22 hour. Soxhlet extraction is achieved in 8 hours.

4.4.6.2 Determination of total benzene extractables (non-volatile vehicle) (styrene-butadiene resin plus chlorinated paraffins 40 percent and 70 percent in 1:1 ratio)

Weigh 50 to 55g. of well-mixed sample to the nearest mg., using a weighing bottle and wash with benzene into a supercentrifuge bowl of 300 ml capacity. Mix well and dilute with benzene to approximately 80 percent of capacity of the bowl. Benzene may be easily lost during centrifugation Make certain that the gasket is in good condition and the cover sealed tightly. Centrifuge for 1/2 hour at 50,000 rpm.

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Cool the bowl externally before opening. Decant the liquid into a tared 250-ml beaker containing three Berl saddles (as boiling stones) and evaporate to dryness on a steam bath. Extract the material remaining in the bowl by thoroughly mixing with benzene and proceeding as before. Decant into a second tared 250-ml beaker containing three Berl saddles and evaporate to dryness. Generally, the first extract is free of pigment and the second extract has a small amount. The presence of pigment in the decantate is indicated by a cloudy, benzene solution.

When pigment is present in either extract, remove it by the following procedure. Evaporate the benzene extract to dryness on the steam bath, take up in benzene and transfer to two 100-ml conical -shaped centrifuge tubes. Centrifuge at 2000 rpm for 45 minutes and decant into the original beaker and again evaporate on the steam bath. If both extracts contain pigment, centrifuge the more concentrated extract first. Use the same centrifuge tubes for the second extract to eliminate possible mechanical loss.

Dry beakers on the steam bath until only a faint odor of solvent remains. (By leaving a slight amount of solvent in the extract, the extract assumes a honeycomb structure upon vacuum drying and is therefore more easily dried). Dry beakers in a vacuum oven at 60 - 70°C and 30 inc. of mercury to constant weight (approximately 32 hours). Weigh beakers and calculate total weight of extractables.

$$\frac{\text{total weight of extracts} \times 100}{\text{weight of sample}} = \text{Percent total benzene extractables.}$$

4.4.6.3 Calculation of resin content.

Percent Styrene-butadiene resin in paint sample = Percent total benzene extractables

4.4.8 Identification of resin from vehicle. Apply a few drops of the benzene extract (see 4.4.6.2) to a sodium chloride disc and as seen by a trial infrared spectrum, dry disc in vacuum oven overnight at 60°C or until the benzene is completely removed as shown by absence of an absorption band at 14.8. (The presence of benzene would affect the 10.3 μ/3.28 μ ratio.) Obtain an infrared spectrum by the usual procedure. Compare with that shown in figure 1. There should be a close resemblance. These styrene-butadiene paint resins are of high styrene type. Draw the baselines as illustrated in figure 1 and determine the 10.3 μ/3.28 μ ratio and compare. The ratio is 0.95 for 83 percent styrene to 0.65 for 89 percent styrene. The ratio should be in this range.

PAGE 5

TABLE IV, lines 9 and 10. Delete:

"Aromatic hydrocarbon solvent[3/]	225.9
Mineral spirits	225.9"

And substitute:

"Rule 66 replacement solvent	451.8"

Footnote[3/]. Delete in entirety.

PAGE 6

Figure 1. Delete caption and substitute the following:

"Infrared spectrum of styrene-butadiene resin, with chlorinated 40 percent and chlorinated paraffin 70 percent in the proportion 2:1:1 based on formulation in Table IV.

Figure 2. Delete.

CIVIL AGENCIES INTEREST:

DOT - SMS, RDS
COMMERCE - NBS
INTERIOR - BPA, BOR
HEW - NIH
VA - DMS
GSA - FSS

DC 72-9320

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